TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Jung Han Choi	Email	jung-han.choi@hhi.fraunhofer.de
Institute Name	Fraunhofer Herinrich-Hertz Institute (HHI)			
Address	Einsteinufer 37, 10587 Berlin, Germany			

2. Technology Overview

Title of Proposed Technology	D-Band Wide-Area Imaging Transceiver System Using multi-layer Glass Interposer for Large-Scale Non-Destructive Testing			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Information & Communication		Electrical · Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Satellite/Radio Wave	Electromagnetic Wave Devices	Measuring Instruments	Safety Monitoring/Diagnosis Control Instruments
Concept of Proposed Technology	The proposed technology represents a next-generation approach to D-band (110-170 GHz) millimeter-wave transceiver integration through glass interposer-based heterogeneous packaging. ● **Key Innovations** 1. Ultra-low-loss Integrated Transceiver: ✓ Monolithic D-band CMOS radar transmitter: 110-170 GHz, EIRP >10 dBm ✓ Large-scale receiver array IC: NEP < 10 pW/Hz ✓ Ultra-low-loss multi-layer glass interposer ✓ Ultra-compact and low-cost CMOS transceiver 2. Wide-Area Imaging System Configuration: ✓ 3D heterogeneous integration of ICs with glass interposer, LTCC, and antenna-on-package (AoP) ✓ Sub-wavelength spatial resolution: <2 mm at 150 GHz			

Rationale for Joint R&D Project	● **Technical Necessity** - Glass interposer technology for D-band RF applications is still in early development globally; no established industry standards or leading market competitors exist - The transition from flex-line to glass interposer requires specialized manufacturing expertise, precision assembly processes, and comprehensive characterization, not yet available domestically ● **Commercial and Strategic Drivers** - Current non-destructive testing equipment market valued at $11.6B (2024), growing at 12.1% CAGR; THz imaging segment at $146M with 18.4% CAGR - Existing solutions rely on compound semiconductors (GaAs, InP) with < 20% production yield, resulting in 50% higher cost than CMOS equivalents - CMOS-based D-band transceiver platform enables mass production, 50%+ cost reduction, and market disruption in quality control applications (food, automotive, battery, semiconductor packaging inspection) - Sub-millimeter resolution imaging meets critical requirements for post-packaging defect detection in advanced semiconductor manufacturing ● **Development Timeline and Cost Optimization** - Joint R&D reduces total investment cost by an estimated 40-50% through shared infrastructure access and intellectual property licensing - Korean institutions alone cannot achieve glass interposer technology maturity within the project timeline (significant capital investment in fabrication infrastructure required) ● **International Collaboration Benefits** - Secured access to Fraunhofer's proprietary glass substrate fabrication processes and characterization infrastructure - Enhanced global competitiveness and technology visibility through association with a world-class research institute - Accelerated commercialization pathway through European market channels and global certification frameworks
Key Words	1. Glass Interposer 2. D-Band Millimeter-Wave CMOS Transceiver 3. Heterogeneous 3D Packaging 4. Non-Destructive Testing System

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)

R&D Objectives

- **Primary Objective**

  Develop and demonstrate an integrated D-band wide-area imaging CMOS transceiver system utilizing glass interposer technology, achieving sub-wavelength spatial resolution (<2 mm at 150 GHz) with real-time processing capability and scalable module architecture suitable for industrial non-destructive testing applications.

- **Specific Technical Objectives**

  1. D-band Multi-layer ultra-fine pitch glass Interposer design
     - ✓ Design and fabricate multi-layer glass substrates with optimized D-band transmission characteristics
     - ✓ Implement through-glass vias (TGV) and multi-RDL enabling flip-chip IC integration
  2. Monolithic D-Band CMOS Transceiver IC
     - ✓ TX CMOS IC: 110-170 GHz, > 10 dBm output power
     - ✓ Implement a wide-array receiver IC: 1×2 to 4×4 scalable configurations
     - ✓ Integrate synchronization and signal combining circuitry on chip
  3. System-Level Imaging and Signal Processing
     - ✓ Implement an image reconstruction algorithm fusing transmission/reflection characteristics
     - ✓ Achieve sub-wavelength spatial resolution through phase-precision detection and neural network post-processing
     - ✓ Demonstrate <2 mm lateral resolution at 150 GHz center frequency

R&D Target Specifications

Frequency Band	110-170 GHz (D-band)
TX Output (EIRP)	>10 dBm
RX Noise Equivalent Power (NEP)	<10 pW/Hz
Spatial Resolution (Lateral)	2 mm @ 150 GHz
Field-of-View (Imaging Area)	>500 mm diameter circular region

R&D Plan

- **Phase 1: Feasibility Verification (6 Months)**
  - ■ Objective: Demonstrate proof-of-concept for core building blocks, including D-band CMOS Tx/Rx architectures, glass interposer RF routing, and antenna-on-package concepts.
  - ■ Key Tasks:
    - ✓ Define detailed system requirements and target specifications for the D-band wide-area imaging transceiver (frequency range, EIRP, NEP, resolution, and field-of-view).
    - ✓ Design D-band CMOS transmitter and receiver front-end architectures suitable for integration with glass interposer
    - ✓ Develop preliminary antenna-on-package (AoP) concepts and perform full-wave EM simulations to evaluate link budget and imaging coverage.

- ✓ Establish co-design guidelines between CMOS ICs, glass interposers, LTCC substrates, and packaging processes in collaboration with the foreign partner institute.

■ Expected Outcomes:

- ✓ Verified system architecture and block-level specifications for the D-band imaging platform.
- ✓ Design rules for glass interposer transmission lines and TGV structures at D-band.
- ✓ Jointly agreed process and integration flow for subsequent prototyping in Stage 2.

● **Phase 2: Prototype Development and System Integration (30 Months)**

■ Objective: Develop and validate a TRL 5-level prototype of a D-band wide-area imaging module based on glass-interposer integration, enabling sub-millimeter resolution non-destructive inspection of large-area objects.

■ Key Tasks:

- ✓ Fabricate and characterize D-band CMOS Tx/Rx ICs designed in Stage 1, including wafer-level RF measurements and performance optimization.
- ✓ Fabricate glass interposer substrates with multi-layer RF routing and TGVs, and evaluate insertion loss, isolation, and reliability under D-band operation.
- ✓ Implement flip-chip bonding and heterogeneous integration of CMOS ICs, glass interposers, LTCC boards, and antenna-on-package structures into compact transceiver modules.
- ✓ Develop a scalable module configuration (e.g., 1×2 to 4×4 element tiles)
- ✓ Implement system-level calibration, imaging algorithms, and signal processing for combined transmission and reflection imaging, targeting sub-wavelength spatial resolution.

■ Expected Outcomes:

- ✓ Integrated D-band imaging transceiver prototype module based on glass-interposer heterogeneous packaging
- ✓ Demonstrated wide-area imaging capability with sub-millimeter resolution and improved cost/performance compared to conventional compound semiconductor-based systems.
- ✓ Verified manufacturability of the glass interposer-based RF packaging platform and its scalability toward multi-channel array configurations.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	■ Glass Interposer Technology Leadership ✓ Direct glass substrate design, fabrication process development, and characterization ✓ Lead TGV drilling, RDL patterning, and multilayer integration technology ✓ Oversee RF performance optimization for D-band transmission line design ■ Technical Contributions ✓ Design and fabricate multi-layer glass interposer substrates (4-6 layers) ✓ Implement through-glass via (TGV) and redistribution layer (RDL) processes ✓ Optimize thermal management features for power dissipation ✓ Develop flip-chip bonding interface specifications ■ Deliverables ✓ Glass substrate process specifications and design manual ✓ Fully functional glass substrates for 1-channel, 2-channel, 4-channel configurations ✓ Joint patent applications on glass interposer innovations
Expected R&R of Korean SME Partner	■ Technical Contributions ✓ Lead overall system architecture and integration strategy ✓ D-band CMOS TX/RX IC design and optimization ✓ Signal processing algorithm development (imaging reconstruction, AI/ML implementation) ■ Research Contributions ✓ Design and fabricate receiver array ICs (180-nm CMOS 4-channel detector) ✓ Conduct system-level integration testing and performance validation ✓ Establish imaging benchmarks and validate against non-destructive testing standards ■ Deliverables ✓ System integration and test protocols ✓ Pilot prototype systems ✓ Commercialization roadmap and productization strategy for D-band imaging modules

5. Expected Outcomes/Impact

Expected Outcomes and Impact	● **Technological Achievements** ■ Direct Outcomes ✓ World-first glass interposer implementation for integrated D-band (110-170 GHz) transceiver systems ✓ Monolithic D-Band CMOS transceiver IC ✓ Sub-wavelength imaging resolution: <2 mm @ 150 GHz ■ IP and Knowledge Outcomes ✓ High-impact patent applications ✓ Peer-reviewed publications ✓ Design documentation, technology transfer ● **Commercialization Potential** ■ Market Size and Growth ✓ Non-destructive testing equipment market: $11.6B (2024) ~ $22.9B (2030) at 12.1% CAGR ✓ THz imaging subsegment: $146M (2024) ~ $476M (2030) at 18.4% CAGR ✓ D-band millimeter-wave imaging niche estimated at $50-100M (2025), growing to $300-500M (2030) ■ Market Applications ✓ Food Manufacturing Quality Control ✓ Automotive Component Inspection ✓ Semiconductor Packaging Inspection ● **Industrial and Economic Impact** ■ Domestic Technology Leadership ✓ Establishes Korea as global leader in D-band integrated imaging systems, competing with Fraunhofer HHI, Intel, and compound semiconductor leaders ✓ Enables international value chain development for D-Band CMOS TRX ■ Supply Chain Security ✓ Domestic sourcing of critical D-band component technologies ✓ Reduced reliance on foreign technology dependencies in semiconductor inspection infrastructure ✓ Building blocks for next generation 6G communication and sensing applications

6. References

(1) Brinker, K., Dvorsky, M., Al Qaseer, M. M., & Zoughi, R. (2020). Review of advances in microwave and millimetre-wave NDTE principles and applications. *Philosophical Transactions of the Royal Society A*, 378(2182), 20190279.

(2) Vallius, G., Lisauskas, A., Yuan, H., Knap, W., & Roskos, H. G. (2021). Roadmap of terahertz imaging 2021. *Sensors*, 21(12), 4092.

(3) Glass Interposer Market Outlook 2031. (2023). Market Research Inc., Global Technology Analysis Division.

(4) Yole Development. (2023). Advanced semiconductor packaging technology review. Industrial Research Report.